

#FABCONSQLCON2026

FABCON

Microsoft Fabric
COMMUNITY CONFERENCE

SQLCON

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ATLANTA MARCH 16 - 20, 2026

Microsoft Fabric Security: Best Practices for Network, Data and Governance

Speakers:

- Vengatesh Parasuraman
- Amanjeet Singh

Principal Program Managers, Microsoft Fabric
Customer Advisory Team (CAT)



Before we begin ... aka.ms/fabric-security-workshop

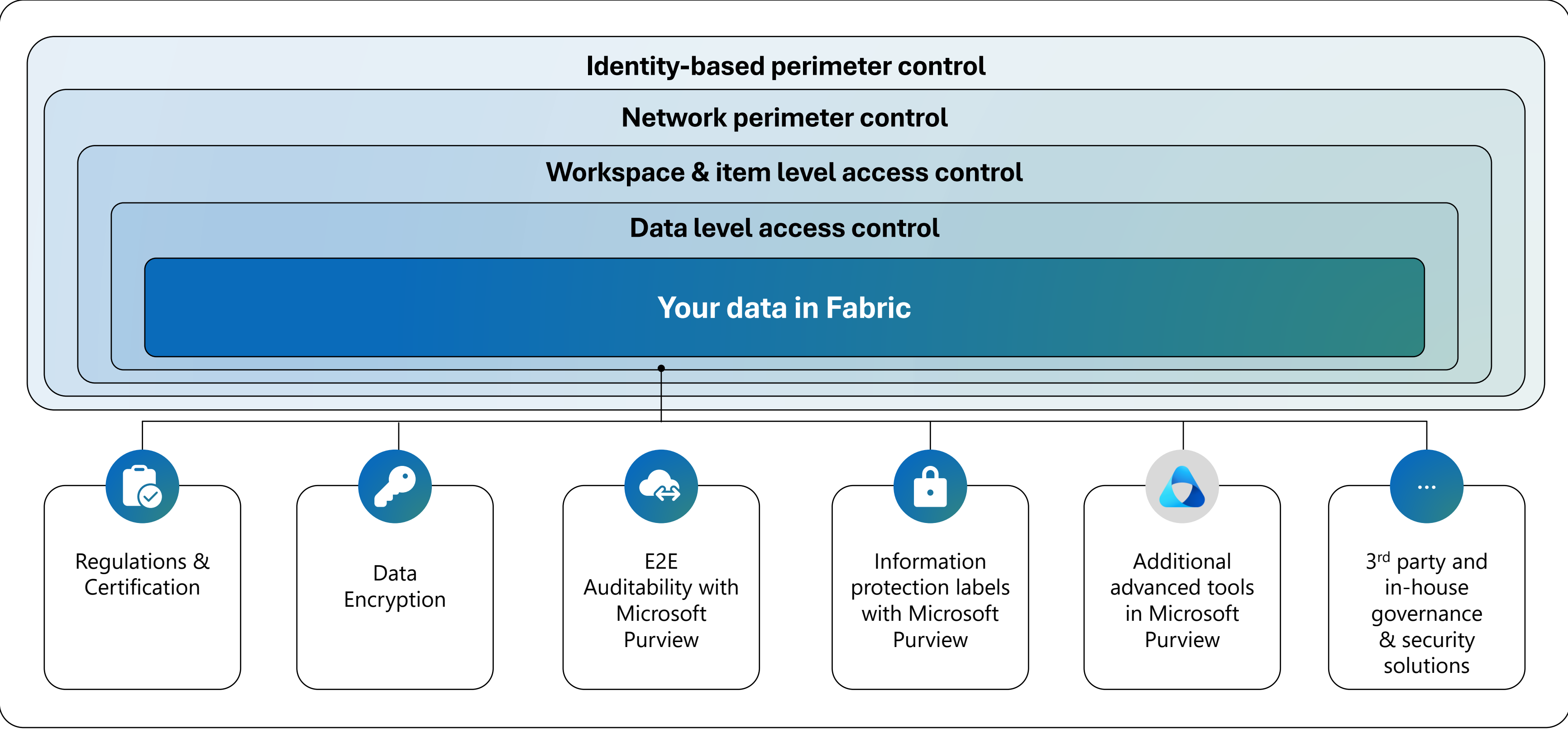
- **Scope & agenda of the workshop**
 - Breaks & lunch
- **Modules**
 - User accounts allocation
 - Workspace layout
 - Azure portal
 - Reminder – this is a shared environment – please DO NOT delete anything and ensure you ONLY approve requests for your user-account i.e. Managed PE approvals.
- **What are we covering today?**
 - Securing inbound access
 - Securing outbound access
 - Securing the control plane
 - Securing the data

Preamble

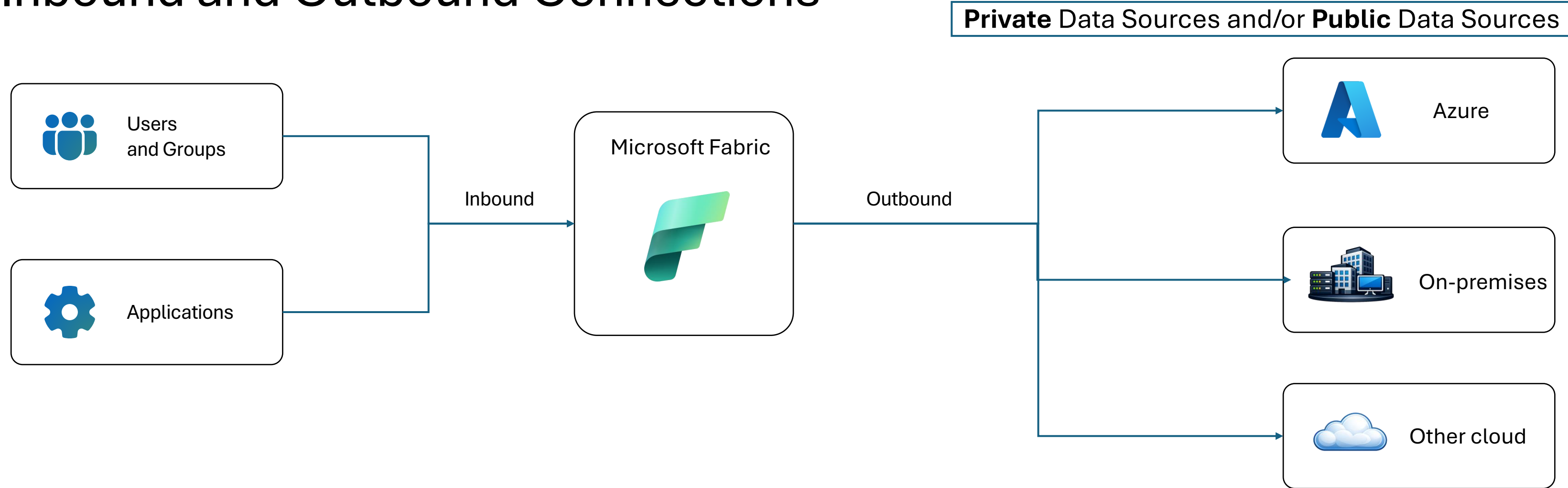
- Microsoft Fabric is built on Zero-Trust principles
 - **Verify explicitly** – Entra integration
 - **Use principle of least privilege** – securing control plane & data plane
 - **Assume breach** – minimize blast radius; segment access; end-to-end encryption; drive detection.
- Defense in Depth: Multiple layers of controls – network, workspace, data and governance.
- Continuous Improvement: Security posture evolves as threats evolve.

[The immutable laws of security | Microsoft Learn](#)

Microsoft Fabric – Defense in depth



Inbound and Outbound Connections



Inbound Connections

Users and applications accessing data in Fabric from their client devices and network

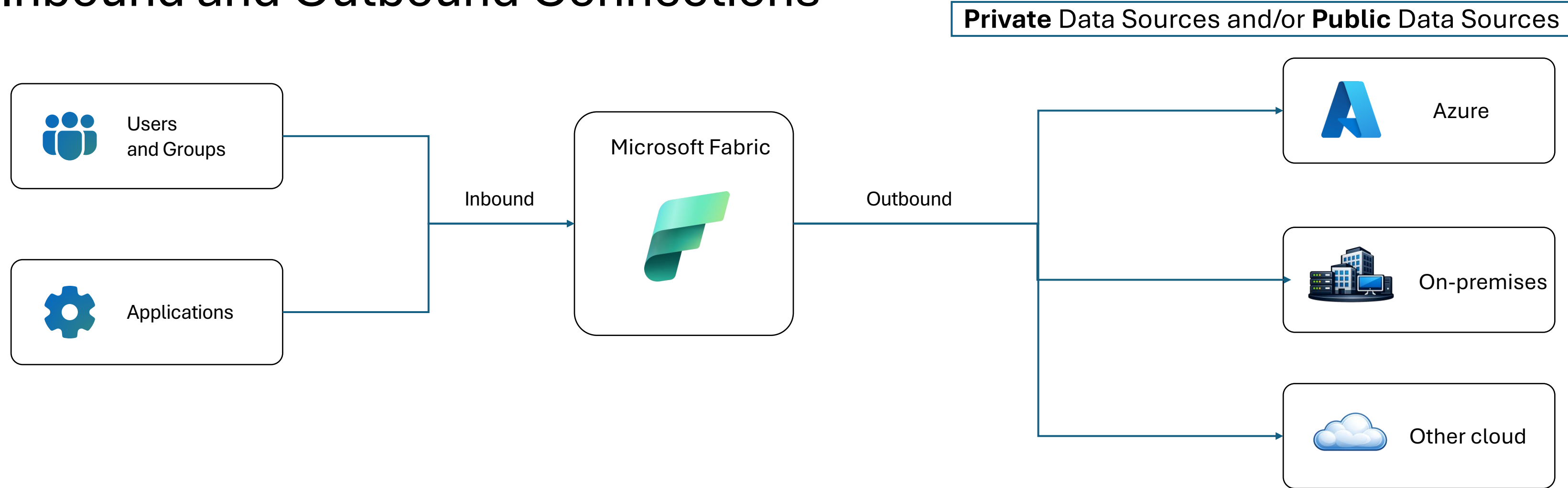
- Connections originate outside of Fabric
- Protected by Entra conditional access, private endpoints, IP Firewall rules

Outbound Connections

Fabric accessing data residing outside such as in Azure, On-premises and other clouds

- Connections originate from Fabric
- Leverages Cloud Connections, Gateways, Managed Private Endpoints and Trusted Workspace Access
- Protected by outbound access protection in Fabric

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Inbound Protection Options

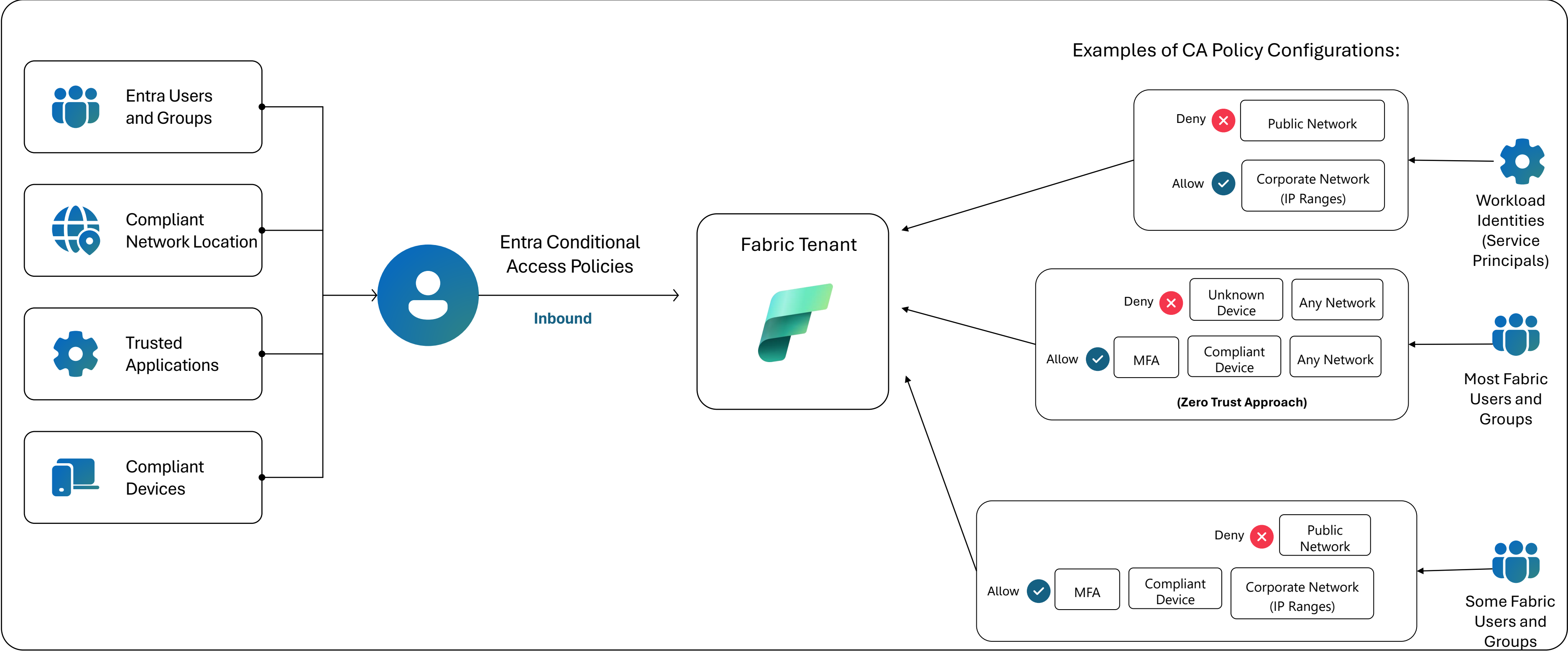
Perimeter Network Security



Zero Trust Approach



Zero Trust for Fabric with Entra Conditional Access Policies



Entra CA and Traditional Network Perimeter Security

What Entra Conditional Access Is

Identity-based perimeter control (Zero Trust)

- Treats **identity as the primary security perimeter**
- Enforces access **at authentication time**
- Uses signals such as:
 - User & group identity
 - Device compliance
 - Location / network signals (IP, country, GPS)
 - Risk & MFA requirements
- Controls **who can access, from where, and under what conditions**
- Protects authentication to all endpoints – public ***and*** private
- ✅ Acts as an **identity-based access perimeter**

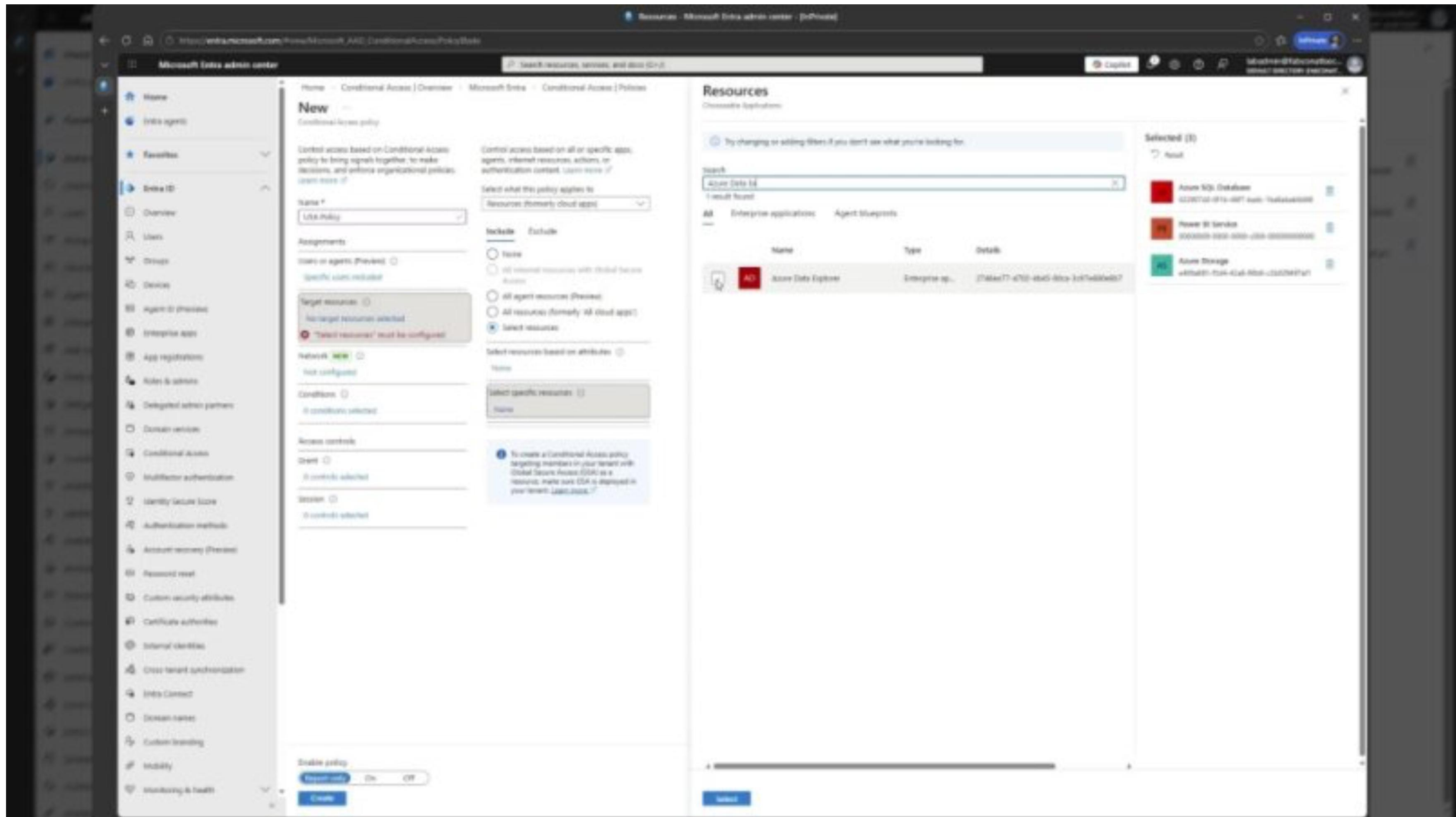
What Entra Conditional Access Is *Not*

Not a traditional network perimeter

- Does **not** isolate services from the public internet
- Does **not** change network routing or endpoints
- Does **not** replace firewalls, Private Link, or VNet isolation
- ❌ Not equivalent to **network perimeter security**

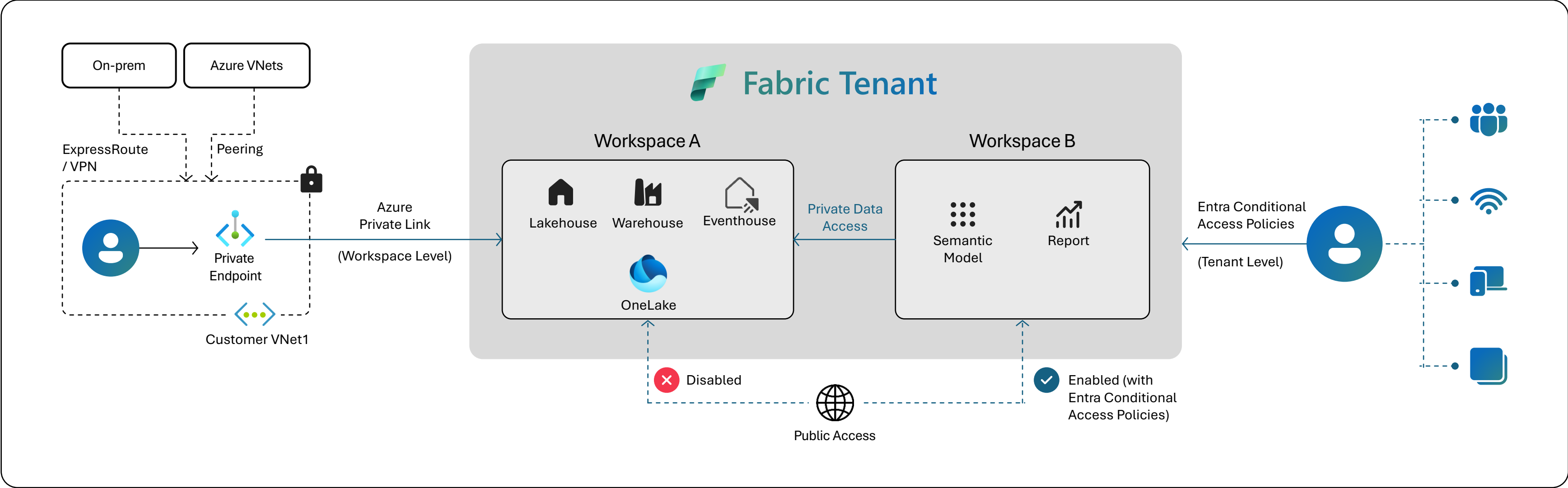
Demo – Configure Entra Conditional Access policy

[Entra CA Demo.mp4](#)



Workspace level Private Link for Fabric

Perimeter Network Security for your workspace



Selected workspaces can be protected using Private Links and closed from public internet.

Create a secure connection between public and private workspaces using private data access.

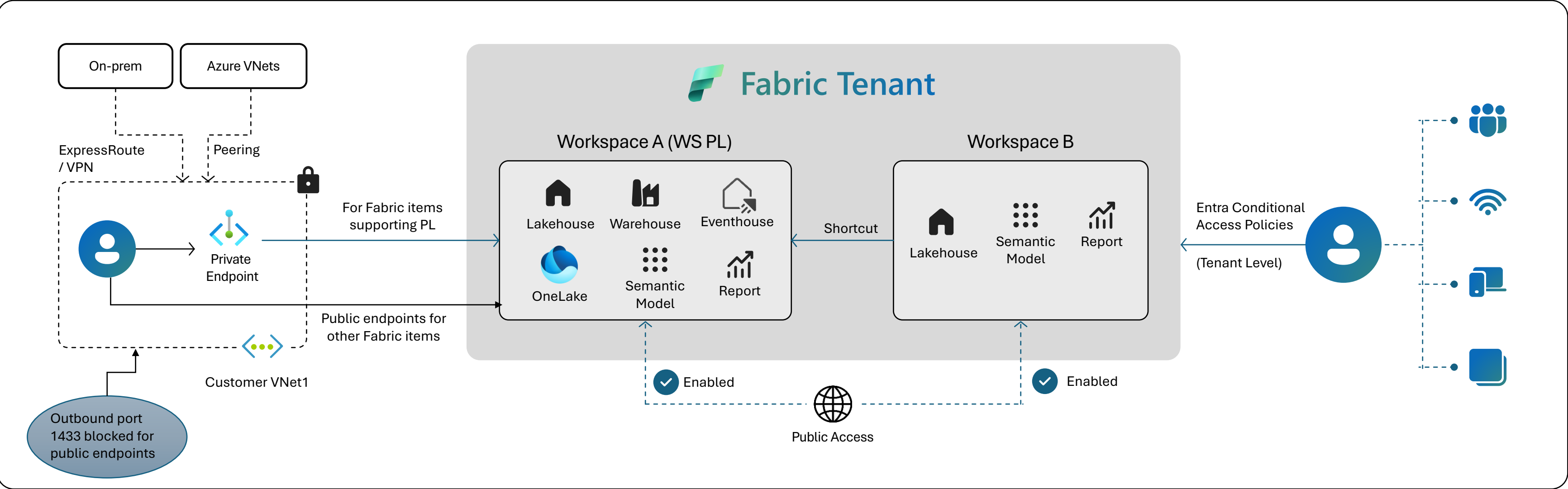
Public workspaces are secured through Entra policies for example to use Power BI.

Considerations and limitations apply.

Hands-On Exercise: Enable Fabric WS PL and configure VNet and PLS on Azure.

WS PL – Combining Private and Public Access

Private Connections on port 1433



Selected workspaces can be protected using Private Links while enabled for public access

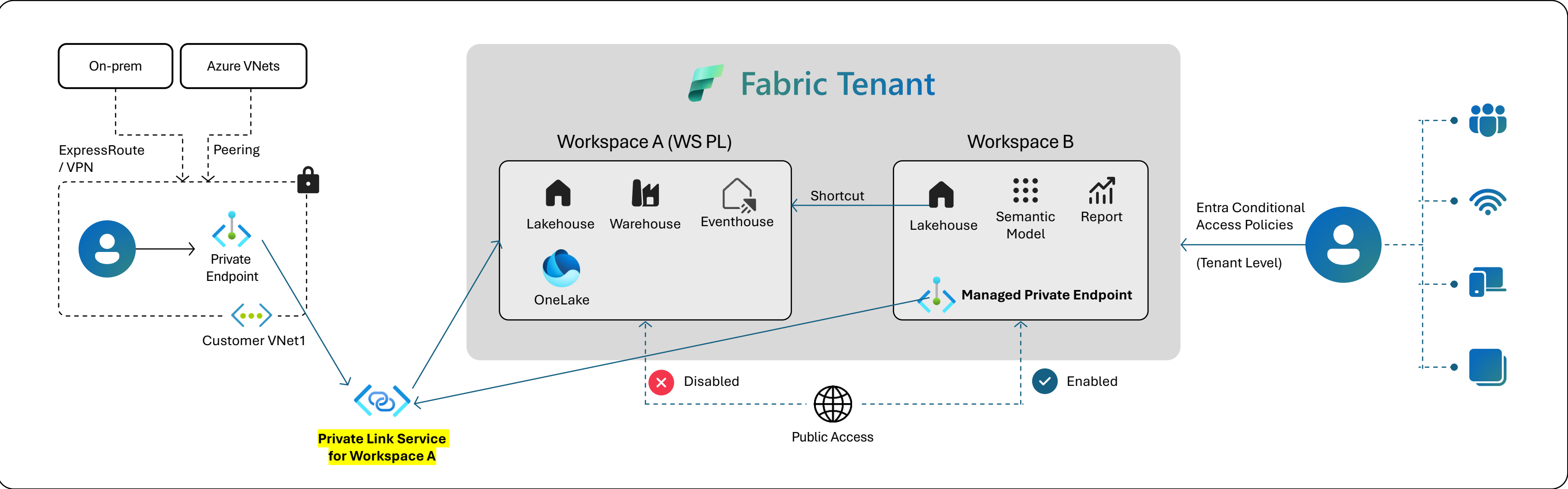
Users from corp. network access via private endpoint (eg: SSMS connecting to SQL Endpoints on port 1433)

Allows authorized users access to Fabric via public endpoints, protected by Entra Conditional Access

DirectLake mode and shortcuts work seamlessly

WS PL – Private Access with Managed Private Endpoints

Direct Lake Mode



Selected workspaces can be protected using Private Links and disabled for public access

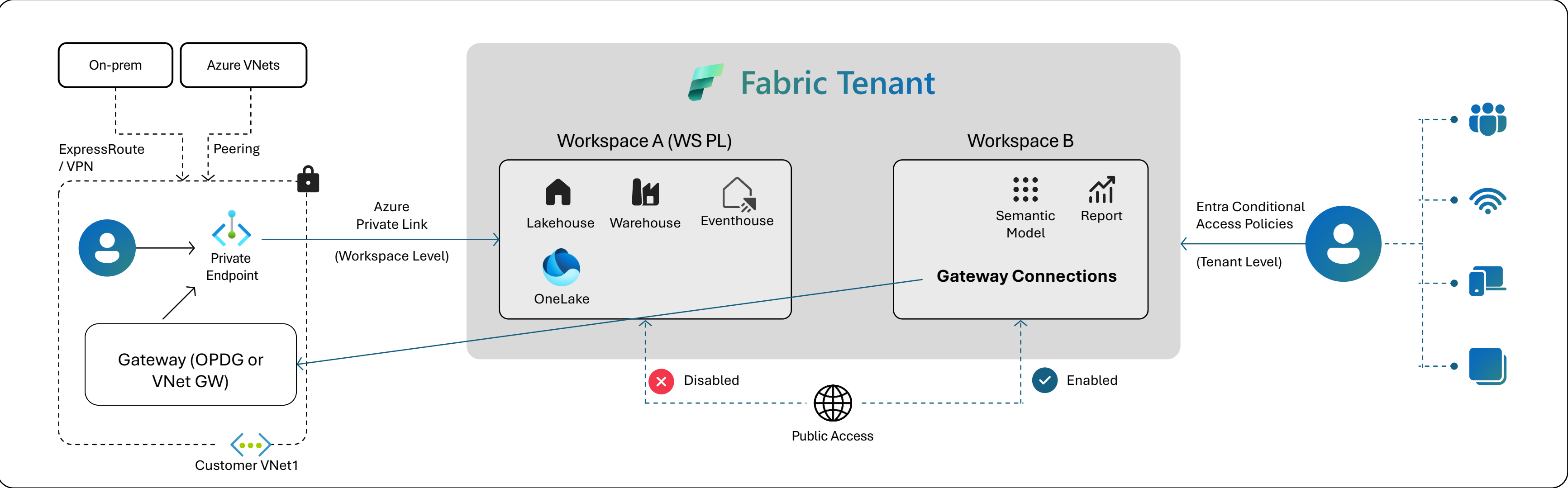
DirectLake mode can be supported by shortcuts via managed private endpoints

This exposes some Lakehouse data on public endpoints

Hence it is recommended to leverage additional security measures such as OneLake Security to minimize data exposure

WS PL – Private Access with Gateways

Import Mode and Direct Query Mode



Selected workspaces can be protected using Private Links and disabled for public access

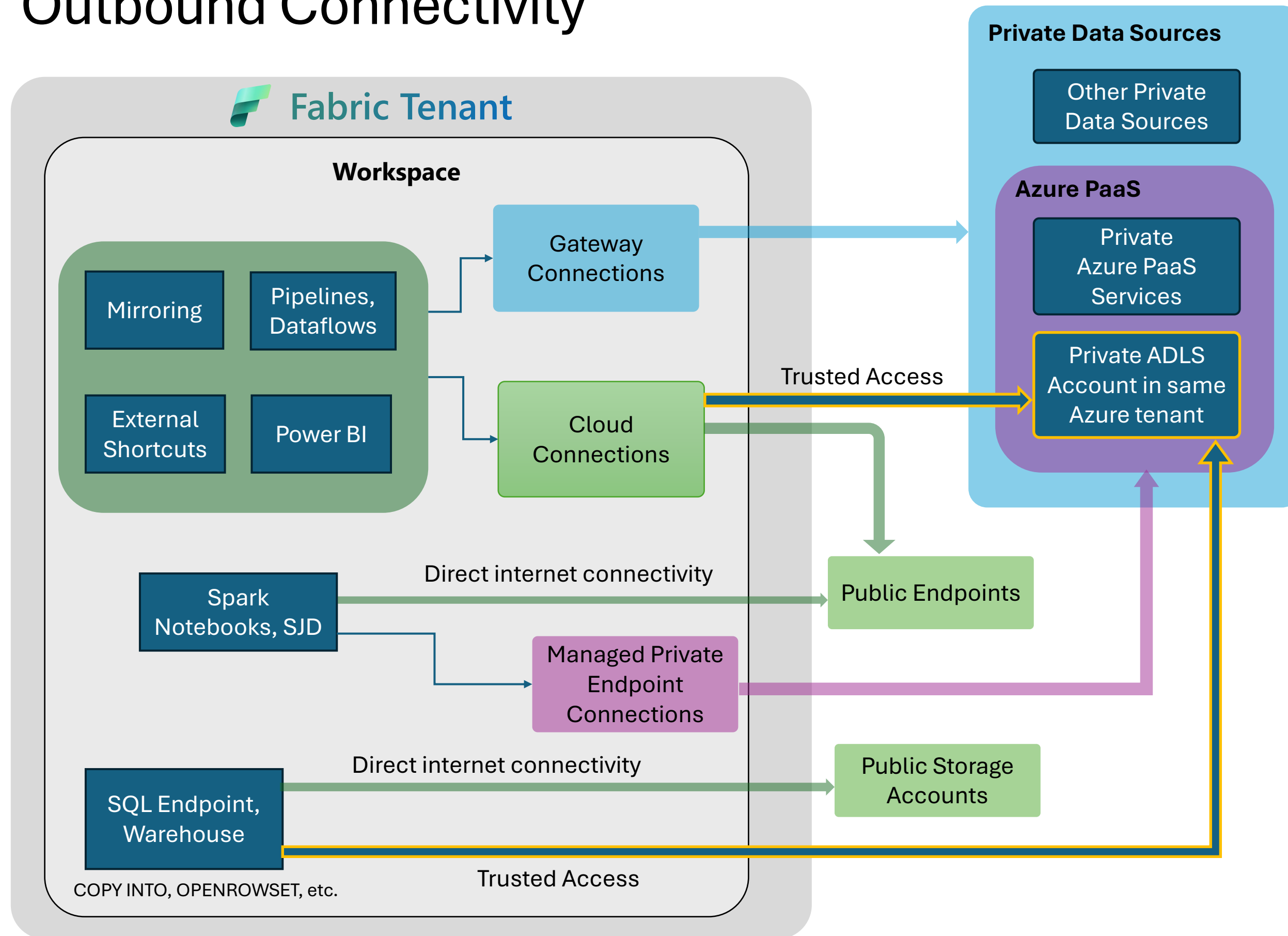
Cross-workspace communications are enabled through gateway connections (OPDG and/or VNet GW)

No DirectLake mode – only import mode and/or Direct Query mode

No Lakehouse in public workspaces

Hands-On Exercise: Cross workspace communication with Managed Private Endpoints

Outbound Connectivity



- Gateway connections are used to connect to any private data source – Azure, on-prem and other clouds

- Cloud connections are generally used to connect to public data sources over public endpoints

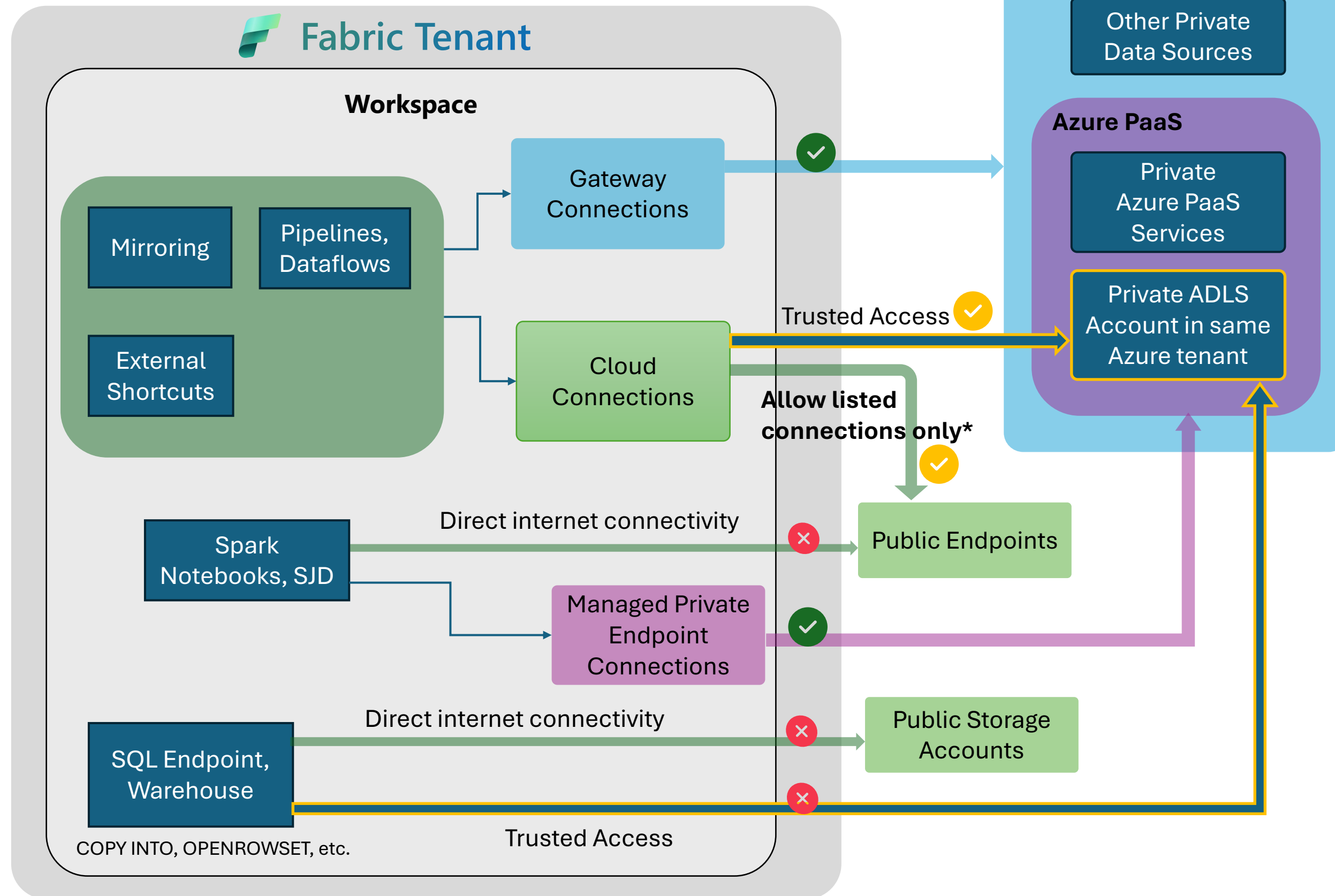
- Only exception is private ADLS in same tenant

- Spark workloads use managed private endpoints to connect to private Azure PaaS services

- Spark workloads also have direct connectivity to internet to access any public endpoints, by default

- SQL workloads connect to public storage accounts directly; for private ADLS accounts, trusted access is used

Workspace Outbound Protection



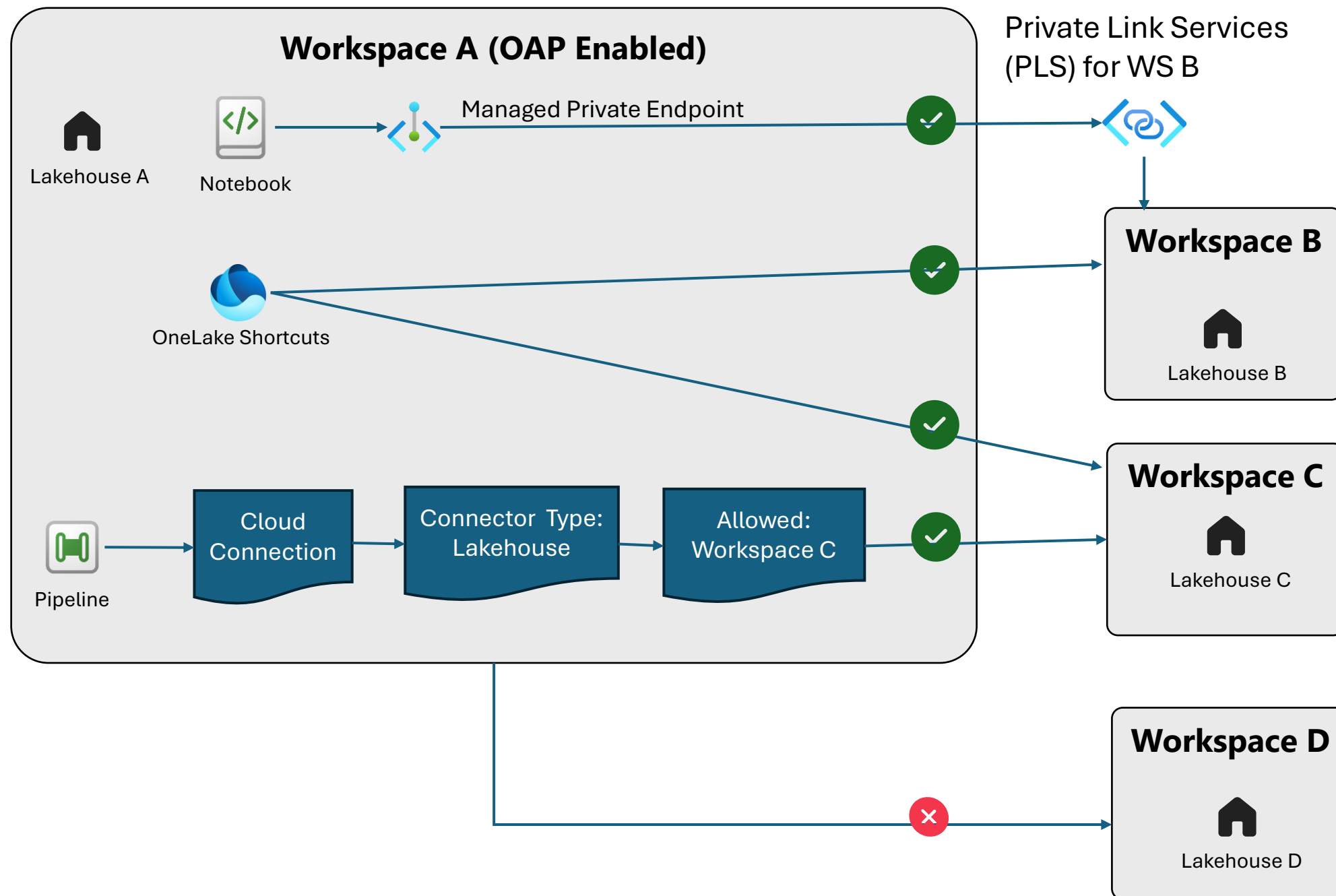
- Gateway connections can be allowed on specific gateways
- Cloud connections can be blocked or allowed to **specific endpoints***
 - Trusted access to ADLS can be allowed to specific ADLS accounts
- Spark workloads **MUST** use managed private endpoints to connect to other Azure PaaS services
- Direct connectivity to internet from spark workloads are **completely** blocked
- SQL workloads cannot connect to any public storage or private ADLS
- Unsupported workloads (such as Power BI, etc.) cannot be created in the workspace

Hands-On Exercise: Enable Outbound Security and setup Managed PE for ADLS gen2.

WS OAP – Cross Workspace Communication

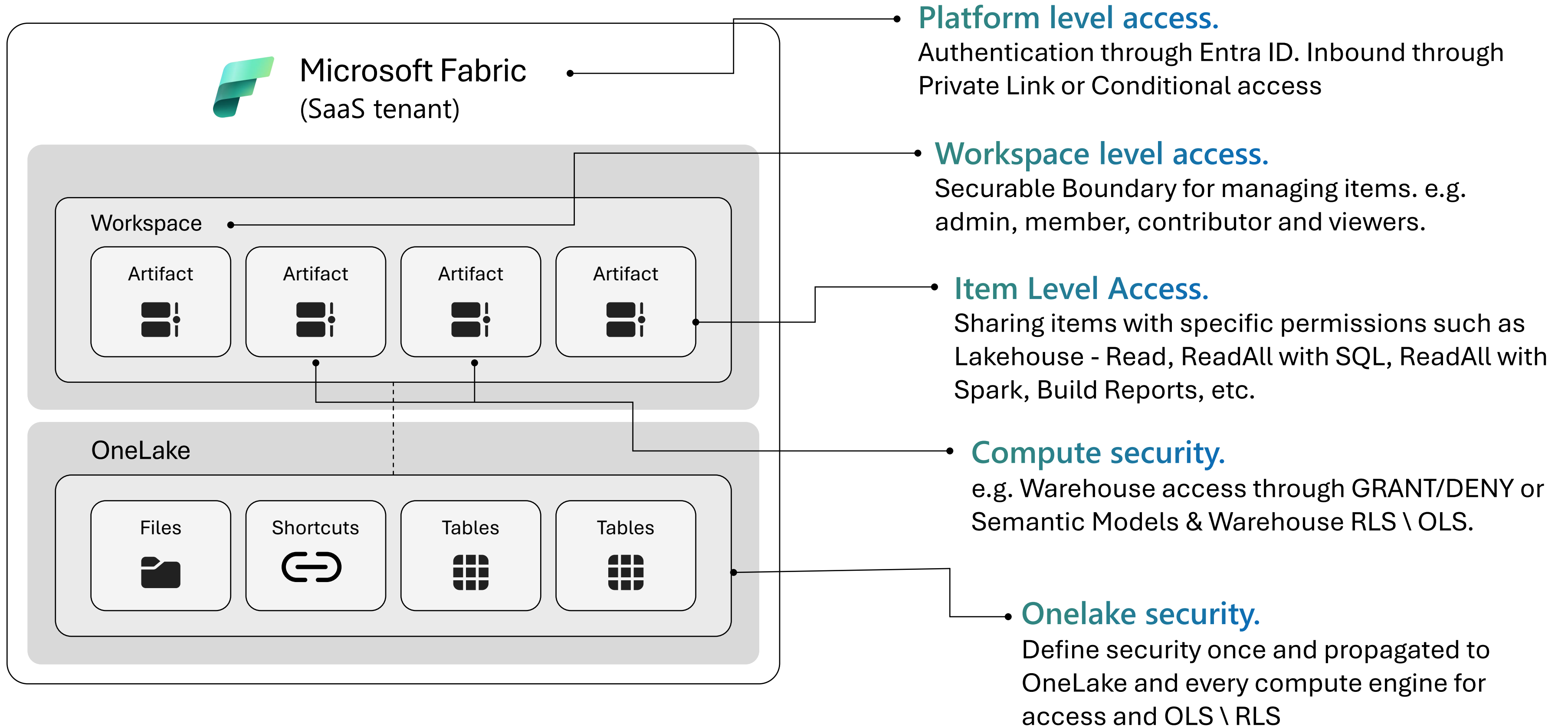
What it means:

1. Outbound connectivity from OAP Enabled Workspace to other Fabric Workspaces are **blocked** by default
2. Notebooks must use Managed Private Endpoints to PLS of other workspaces
3. Pipelines must use connector rules to allow list specific workspaces
4. OneLake shortcuts need either MPE or Connector rule
5. Example: Notebook in WS A can access WS B (and not WS C), pipeline can access WS C (and not WS B), OneLake Shortcuts can be created to both WS B and/or WS C. **No workloads can access WS D.**



Hands-On Exercise: Configure OAP policy for DI Pipeline Copy Activity. You will write data to a LH in a different workspace.

Multiple layers of security and access control



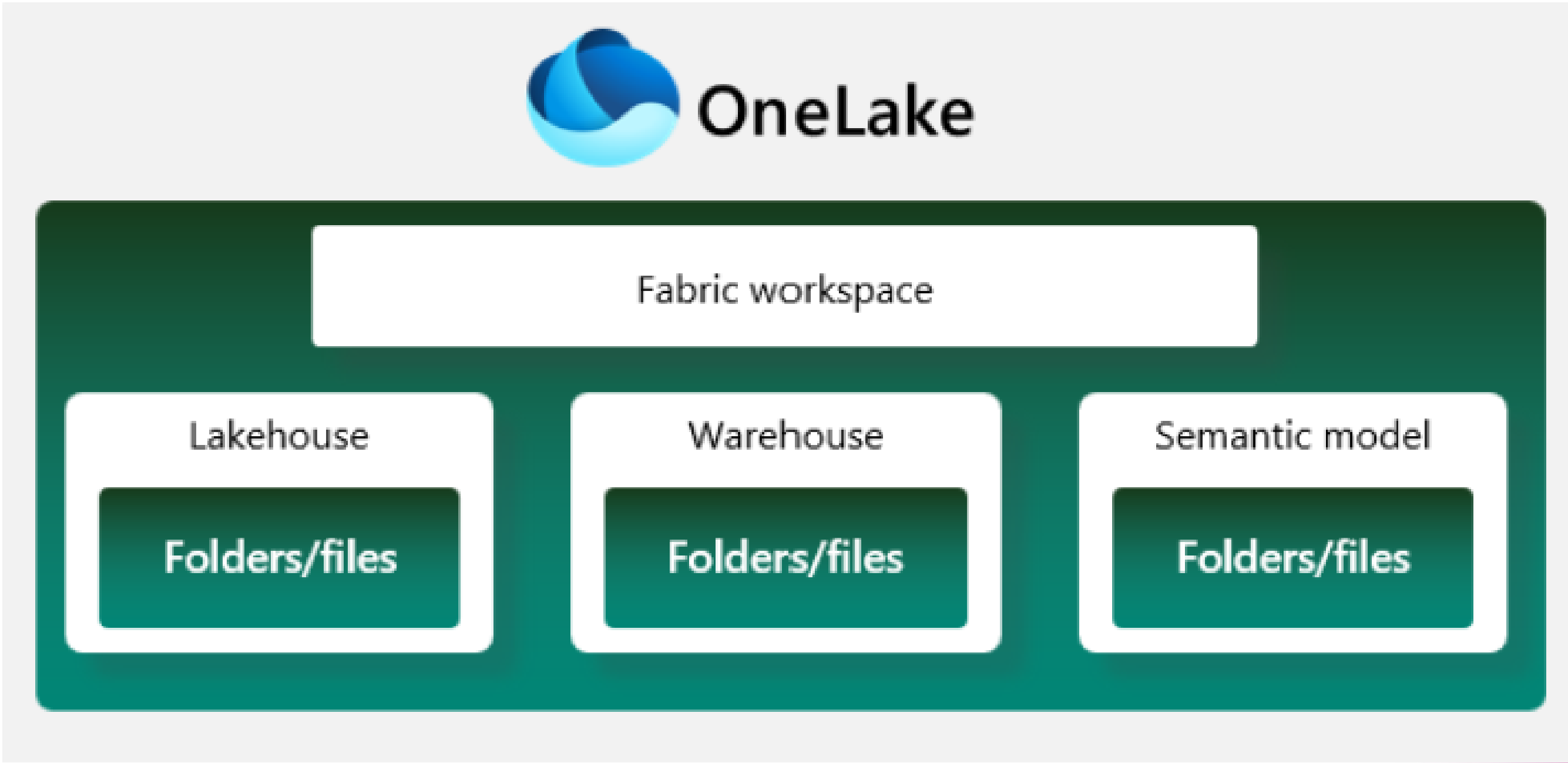
Hands-On Exercise: Assign workspace role to a test user. Test out various item permissions.

Demo: Instructors demonstrate how to use Entra PIM to secure workspace privileged roles (Admin and Member)

What is a Trusted Engine in Microsoft Fabric

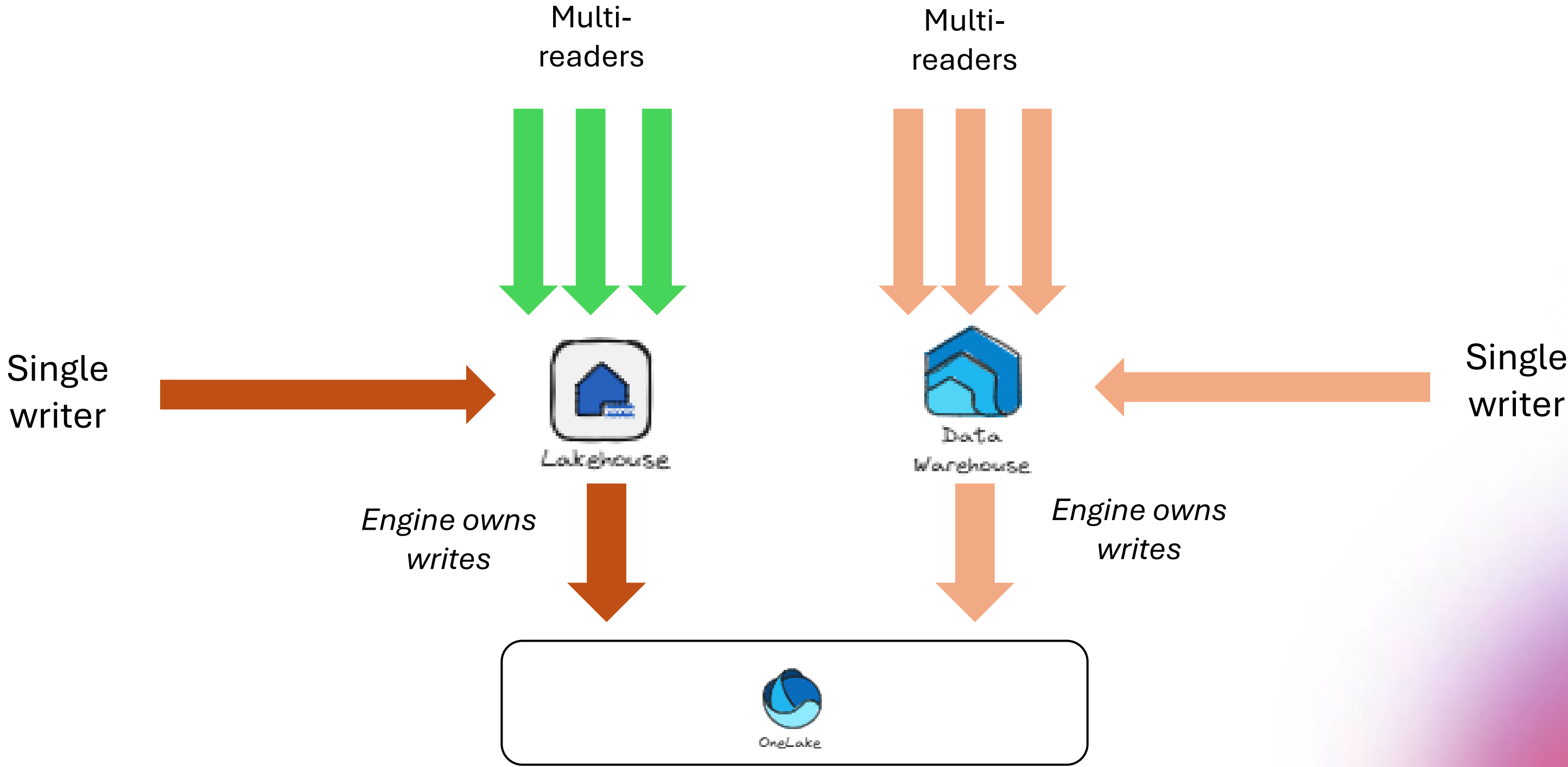
First party / trusted engines

3P engines



Microsoft Fabric – Writers and Readers

Writers own writes (CRUD) and data can be read by many different engines in Fabric (1P) and 3P engines.



OneLake security manages read access

OneLake security provides cross-engine
read access control for OneLake tables.

Writers manage write permissions

- The compute engine writing data (*writer*) owns responsibility of implementing write permissions.
- Exception to this principle is lakehouse. OneLake security is responsible for implementing read and write permission.
- Compute engines have in-built permission model which we plan to continue to leverage for write permissions. Applies to Warehouse, Eventhouse and other engines (where applicable).

Anatomy of a OneLake security role

A role in OneLake security has 3 parts:

- **Members** – user identity, SPNs
- **Scope** – table, folder, schema
- **Permissions** (*and their characteristics*)
 - Grant – table-level, RLS, CLS
 - Deny* – supersedes a *grant* (*work in progress*)

Hands-On Exercise: Apply OneLake security roles and test table-level, RLS and CLS on a pre-provisioned lakehouse.

Demo: View Fabric control plane audit events using Purview

Demo: OneLake Diagnostics

Sound off.
The mic is all yours.
Influence the product roadmap.

Join the Fabric User Panel



Share your feedback directly with our
Fabric product group and researchers.

<https://aka.ms/JoinFabricUserPanel>

Join the SQL User Panel



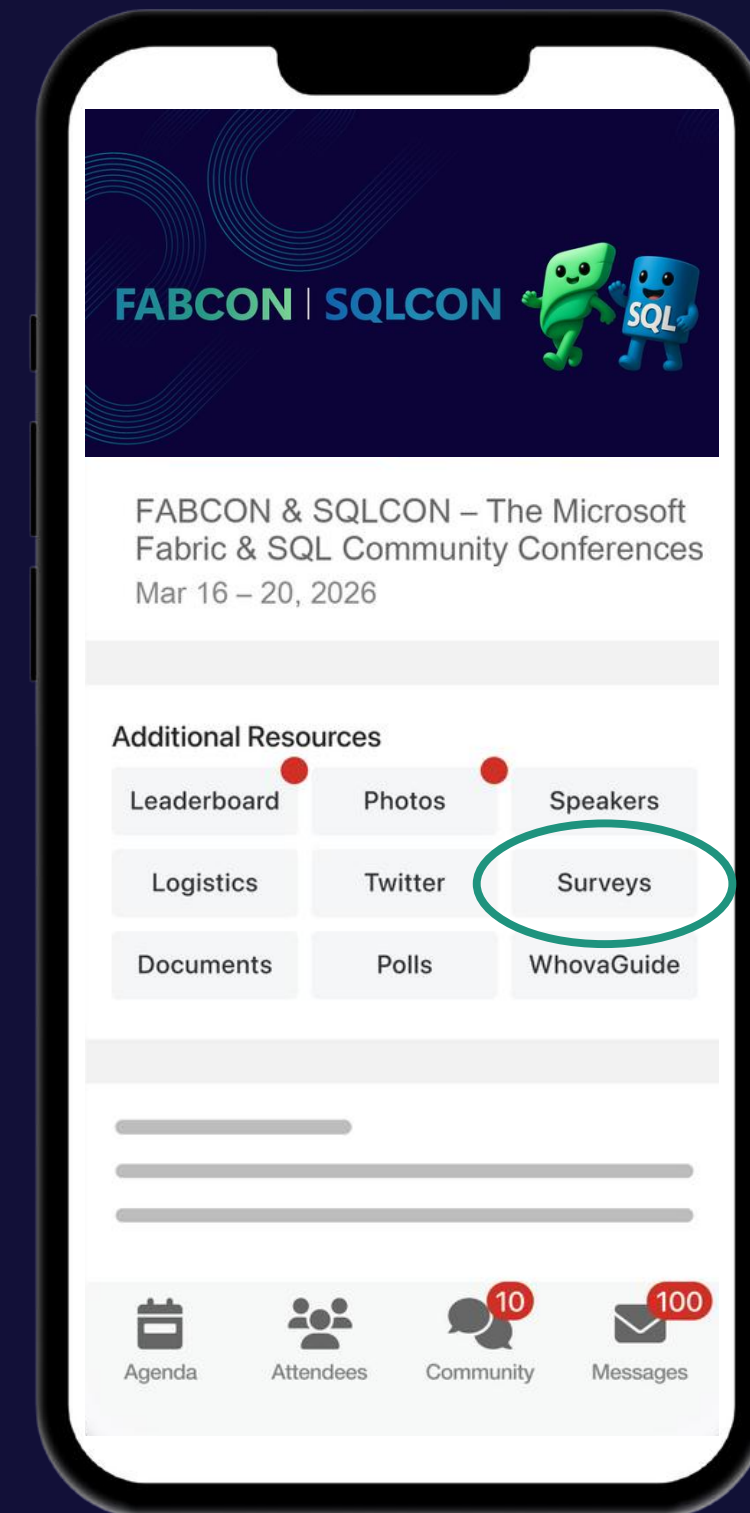
Influence our SQL roadmap and ensure
it meets your real-life needs

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How was the session?



Complete Session Surveys in
Uhova for your chance to WIN
PRIZES!

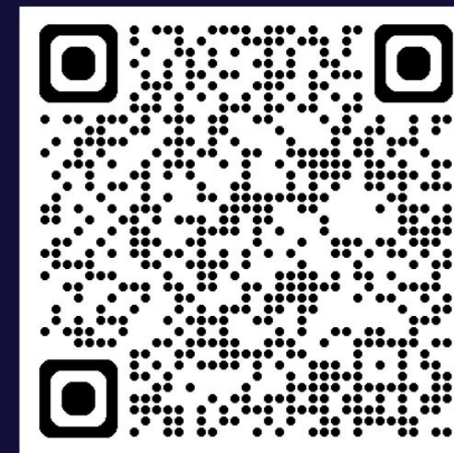


Get Two Fabric Certifications for FREE

Attendees of FABCON can take the Fabric Analytics Engineer or Fabric Data Engineer exam for free. Be part of the 2 fastest growing role-based certifications in Microsoft history.

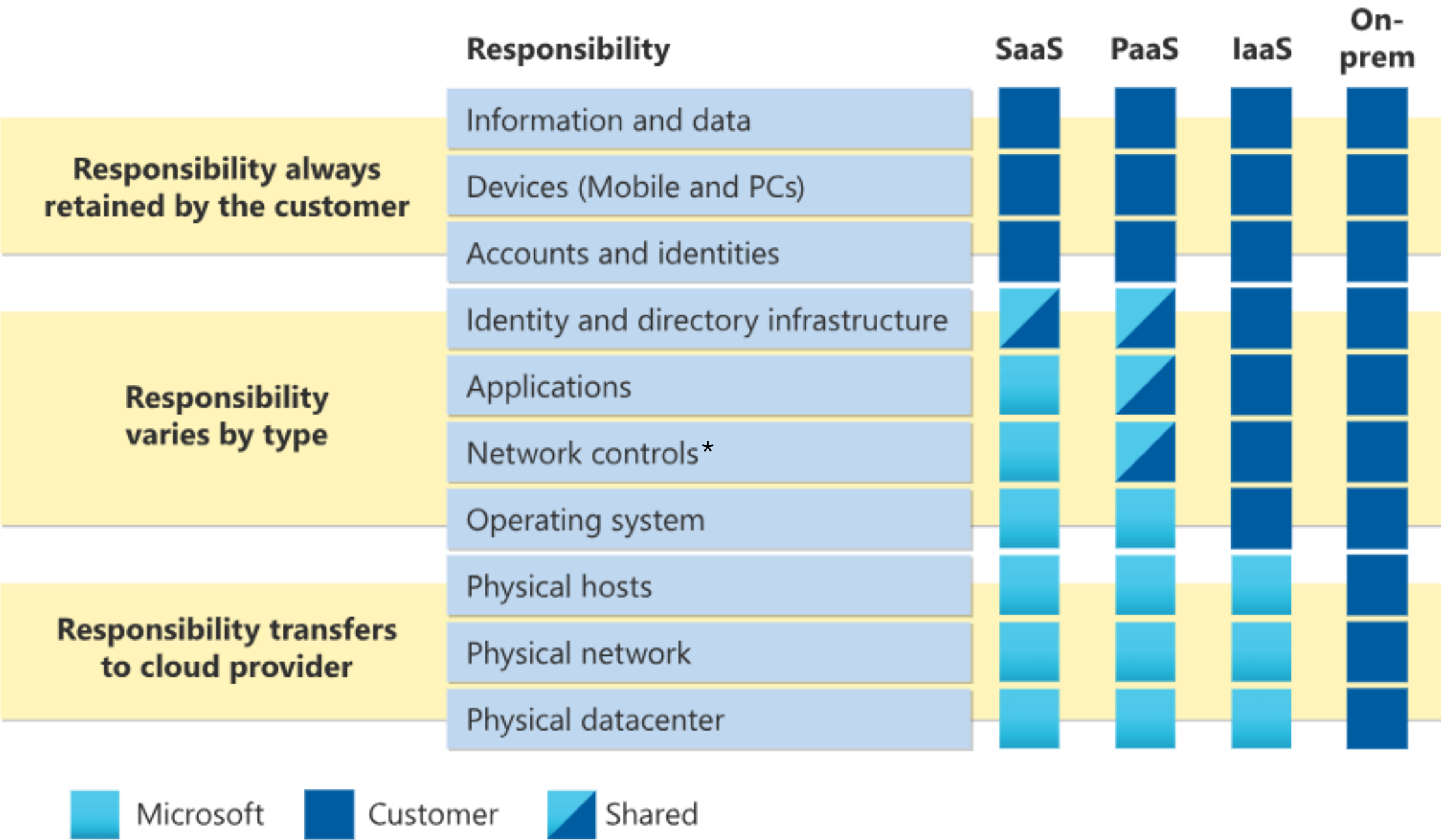
Request your voucher by March 23, 2026.

<https://aka.ms/fabcon/cert100>



Shared Responsibility Model in the Cloud

- Workload responsibilities vary depending on whether workload is hosted as IaaS, PaaS or SaaS.
- Division of responsibility
- Responsibilities you always retain –
 - Data
 - Endpoints
 - Accounts
 - Access management



* Fabric offers additional network controls which can be configured by customers.

Certifications & Attestations

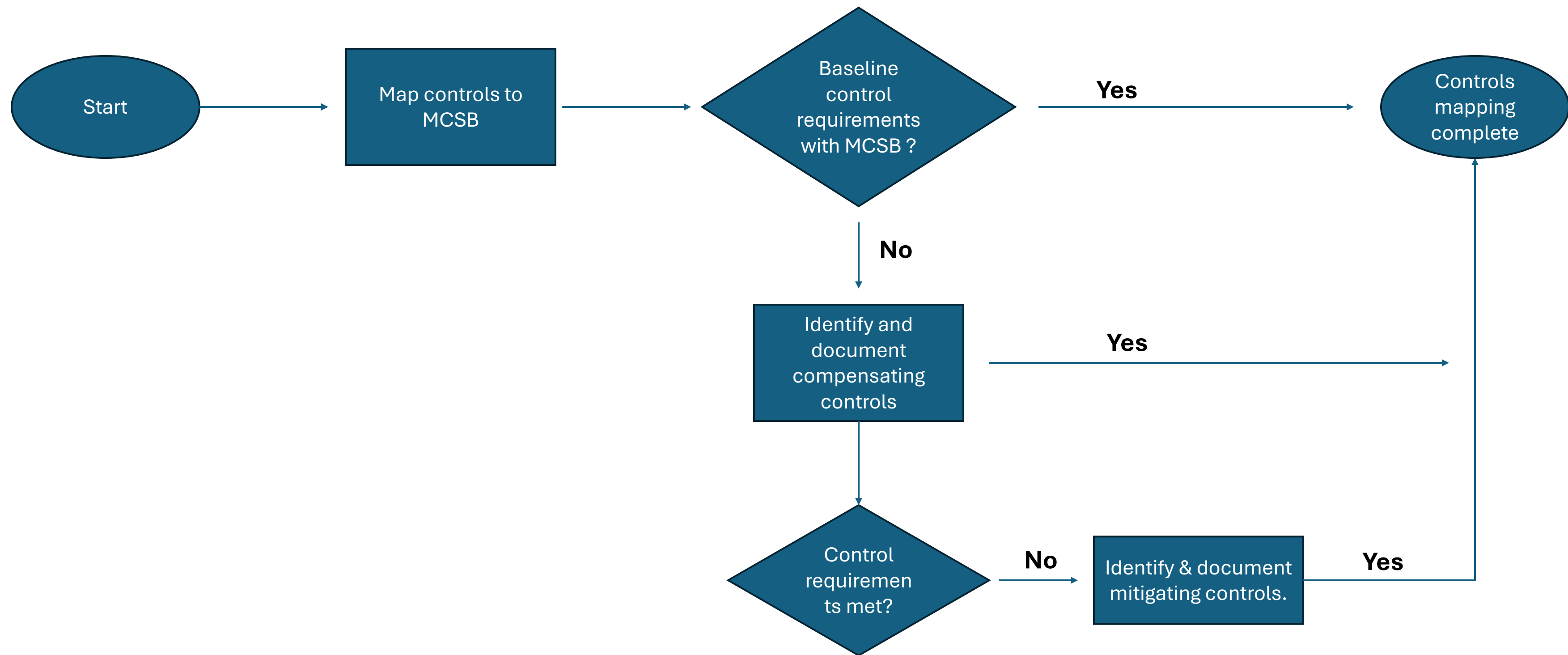
- Independently validate Microsoft's platform-level controls
- Identify Microsoft's responsibilities in the shared responsibility model
- Provide audit evidence for your organisation
- Reduce compliance workload for enterprises
- Enable adoption in regulated industries

Azure Security Baseline for Microsoft Fabric

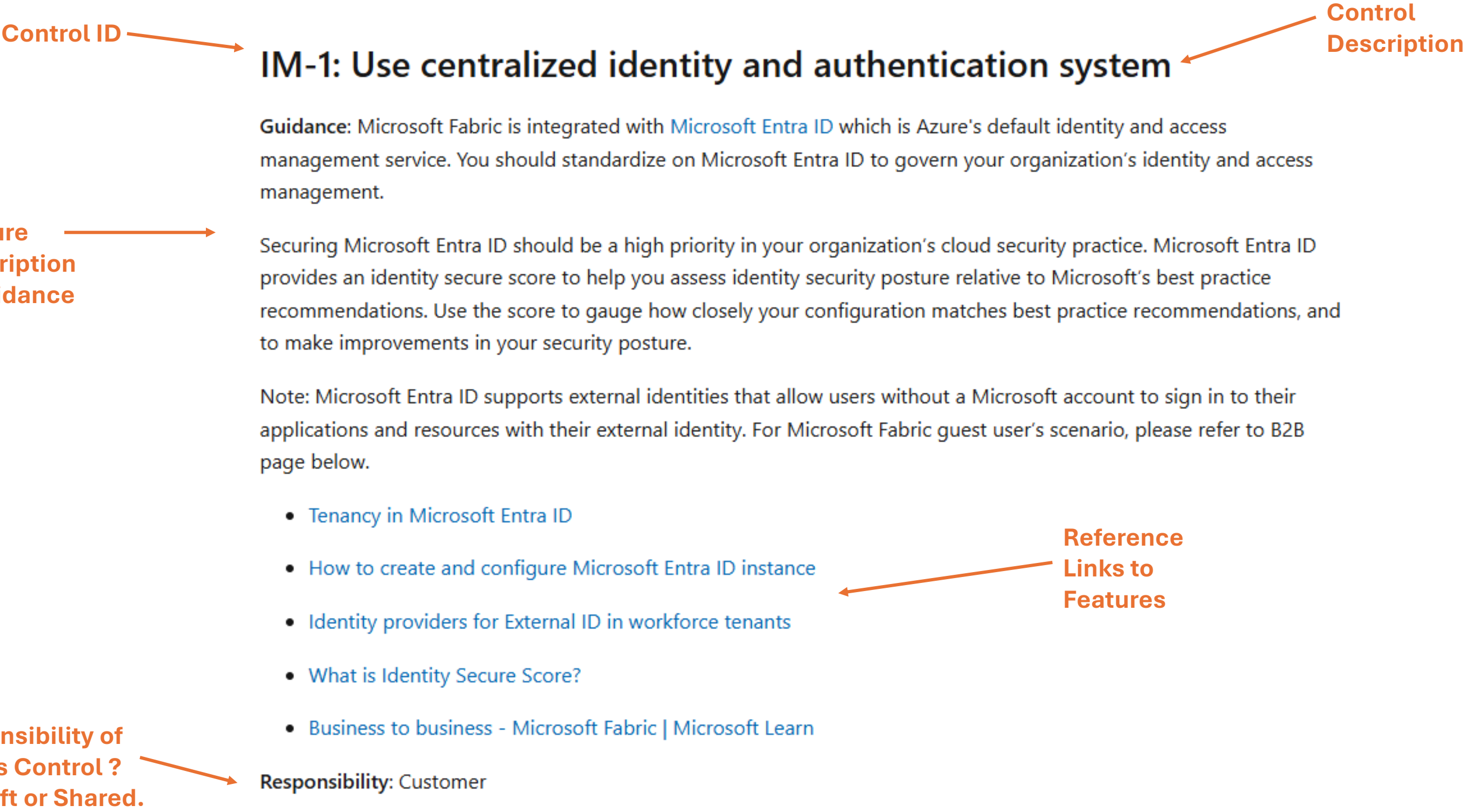
- Certifications validate Microsoft's platform-level controls - but not your tenant's configuration.
- Control effectiveness on *your* side requires periodic review and risk assessment.
- Azure Security Baseline for Microsoft Fabric maps recommended controls to industry standards.
- Baseline helps identify gaps across identity, network, data protection, logging, and monitoring.
- Acts as a practical checklist for reducing risk and strengthening your Fabric security posture.

[Azure security baseline for Microsoft Fabric | Microsoft Learn](#)

Security Controls Mapping



Overview of Security Baseline



[Azure security baseline for Microsoft Fabric | Microsoft Learn](#)

Inbound Network Security

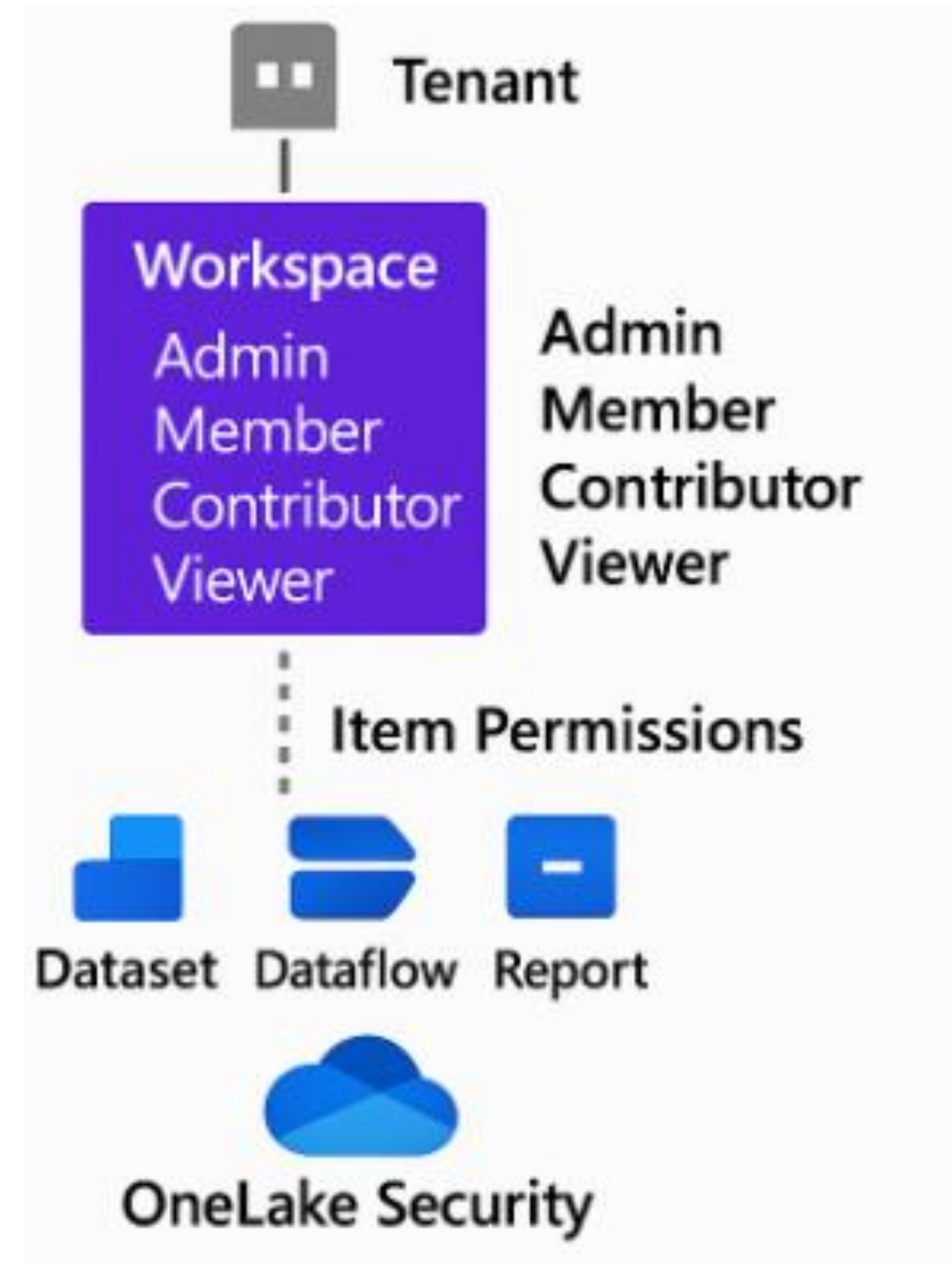
- These features secure inbound connectivity to Microsoft Fabric.
- Scoped at tenant-level and/or workspace-level.
- **Tenant-level inbound protection**
 - Microsoft Entra Conditional Access (Entra CA)
 - Tenant-level private link (TPL)
- **Workspace-level inbound protection**
 - Workspace private link (WS PL)
 - Workspace IP firewall

Outbound Network Security

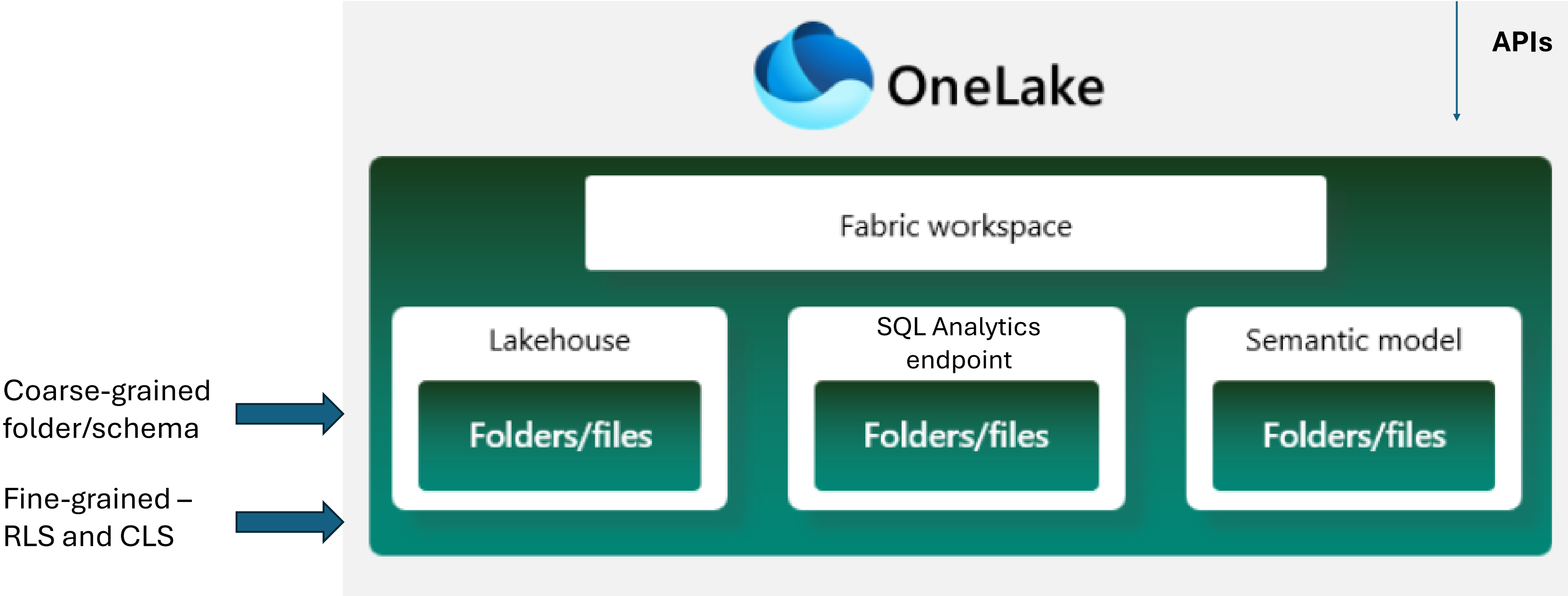
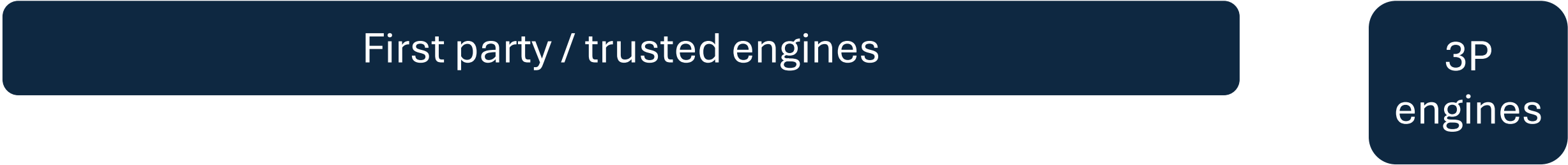
- These features secure outbound connectivity from Microsoft Fabric to external services or other workspaces within a tenant.
- Scoped to a workspace.
- Managed Virtual Networks (VNets) and Managed Private Endpoint (MPE)
- Data Gateways (On-Premises Data Gateway & Managed VNet Gateway)
- Outbound Access Protection
 - For Spark & notebooks via managed private endpoints
 - Data connection rules for Data Integration (DI) and shortcuts
- Trusted workspace access
- Service Tags

Workspace & Item Permissions

- Flexible hierarchical permission model
 - Workspace Roles
 - Item Permissions
 - OneLake security (data plane)



OneLake Security



Tenant level Private Link (TPL) for Fabric

Experience

Advanced networking

▵

Azure Private Link
Enabled for the entire organization

Increase security by allowing people to use a **Private Link** to access your Fabric tenant. Someone will need to finish the set-up process in Azure. If that's not you, grant permission to the right person or group by entering their email. [Learn More](#) | [Set-up instructions](#)

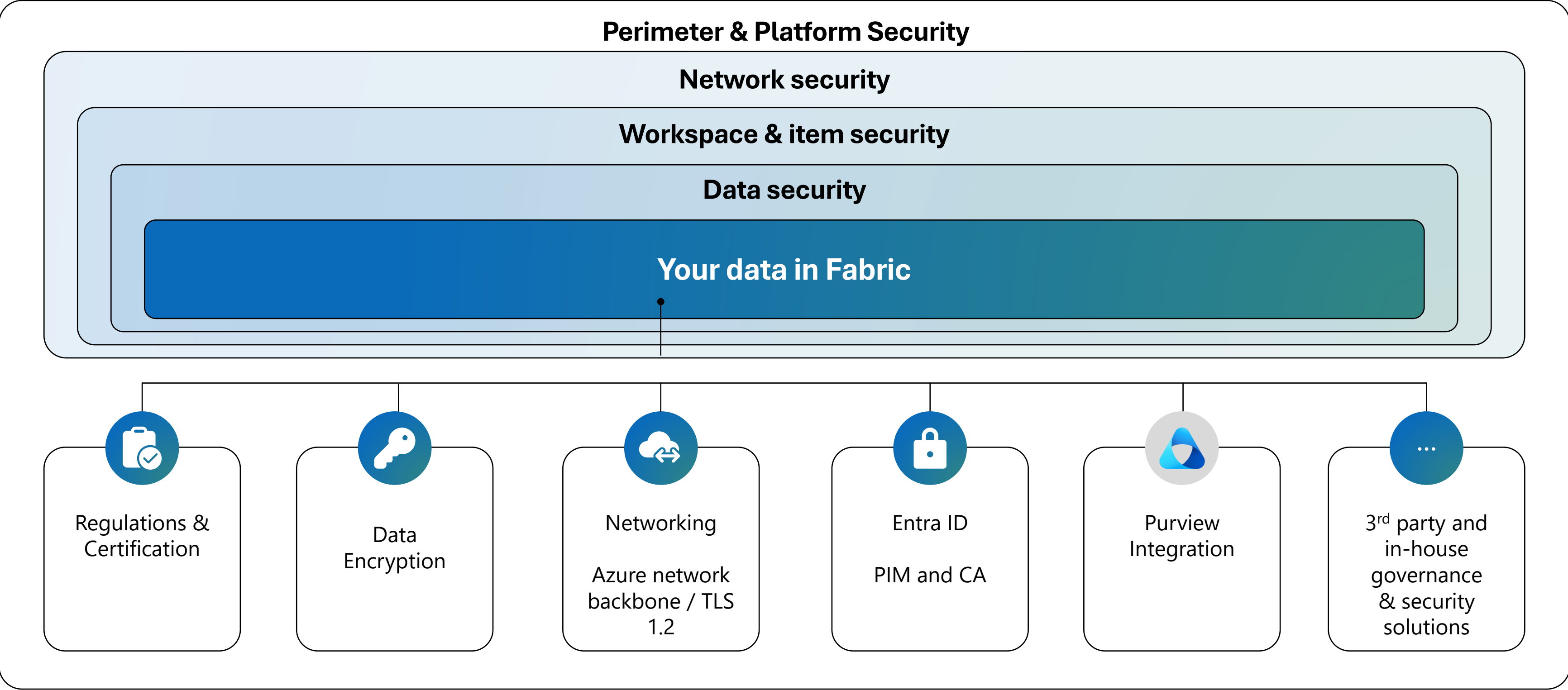
Review the [considerations and limitations](#) section before enabling private endpoints.

Enabled

There are 2 tenant settings in the Fabric admin portal:

- You can either enable tenant PL + enable **Block** Public internet access
- **OR** you can enable tenant PL + disable **Block** public internet access.

Microsoft Fabric – Defense in Depth



Accessing external services/apps from Fabric

How to connect to the data in private network *from* Fabric



Fabric Pipelines (COPY)



Dataflows Gen2



On-Prem Data Gateway



VNet Data Gateway



Airflow



Dedicated Cluster IPs



Mirroring

- Azure SQL DB
- Azure SQL MI
- Snowflake



On-Prem Data Gateway VNet Data Gateway

- Google Big Query
- Oracle
- SQL Server



On-Prem Data Gateway

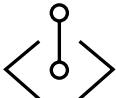
- PostgreSQL



VNet Data Gateway



Fabric Spark



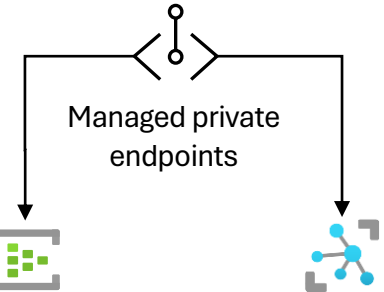
Managed private endpoints



Private Link Services



Fabric Event Streams



Managed private endpoints

Azure Event Hub Azure IoT Hub

(Preview Shortly)



VNet Injection

kafka Other Services



Shortcut to ADLS Gen2



Fabric Pipelines (COPY)



COPY INTO DW



Trusted Workspace Access



OneLake Shortcuts



Amazon S3



GCS



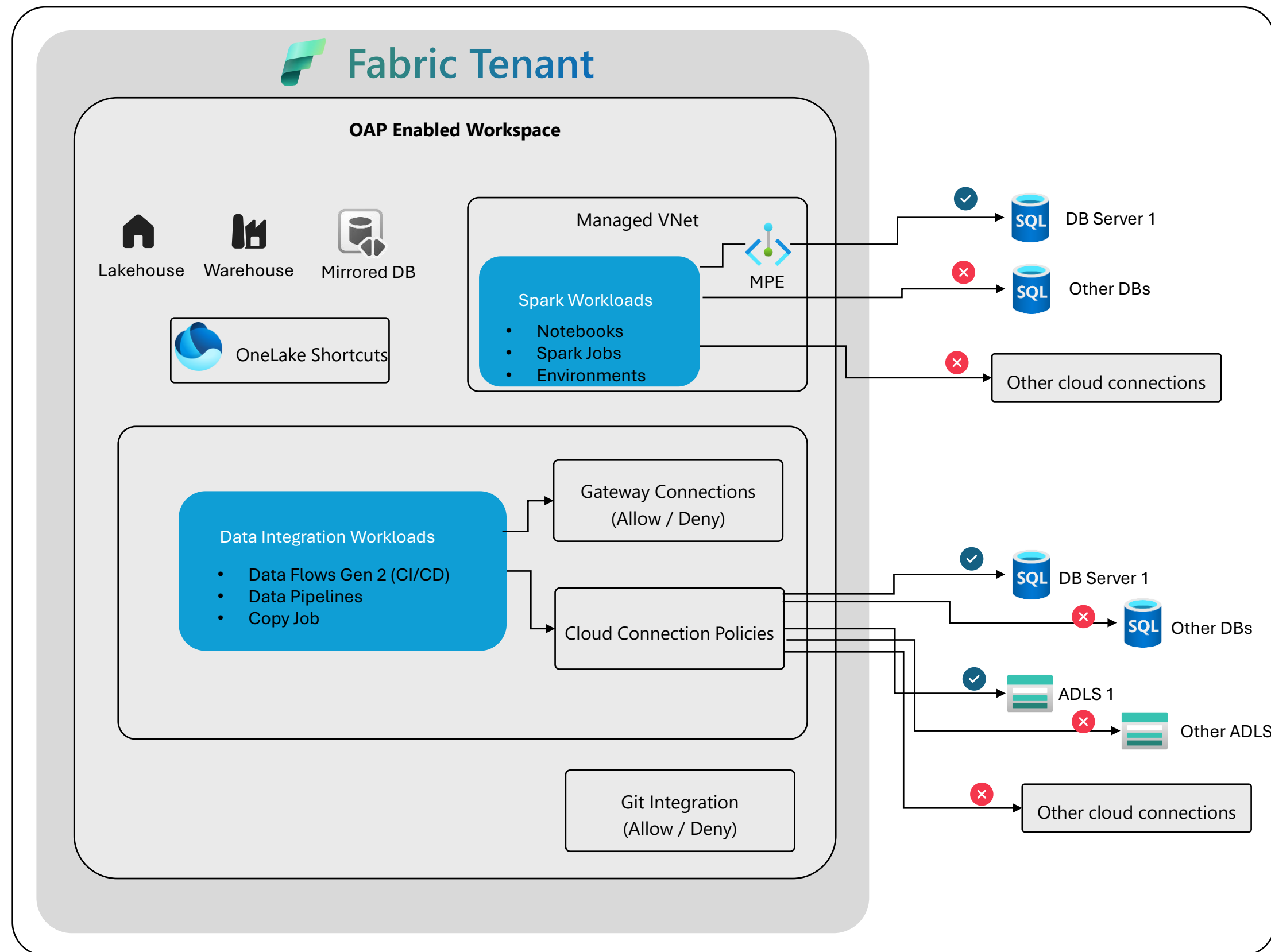
On-prem S3 compatible



On-Prem Data Gateway

WS OAP for Spark and Data Integration

Outbound Access Protection

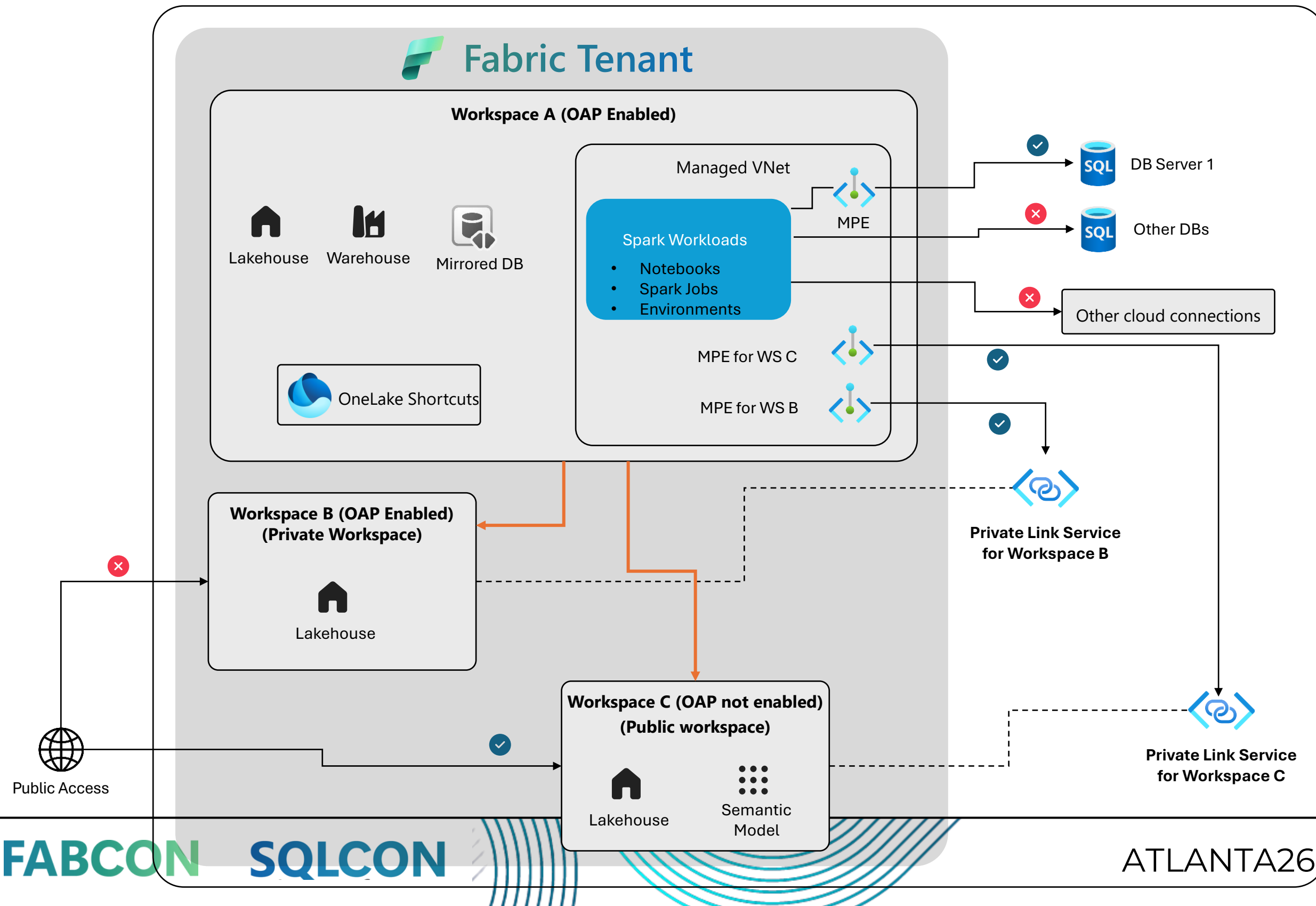


What it means:

1. Allow outbound connections from the Fabric Workspace only to authorized destinations
2. Spark workloads such as notebooks will use Managed Private Endpoints to connect
3. Data Integration workloads such as pipelines will use cloud connection policies to allow-list approved destinations
4. Gateway Connections and Git Integrations can be allowed/denied
5. All other outbound connections to public internet are blocked

WS OAP – Cross Workspace Communication

OAP with Managed Private Endpoints



What it means:

1. Outbound connectivity from OAP Enabled Workspace to other Fabric Workspaces are **blocked** by default
2. Create Private Link Service for each Workspace in Azure
3. Create Managed Private Endpoints to other workspaces in OAP Enabled Workspace and approve in Azure
4. Outbound connectivity from OAP Enabled Workspaces is allowed only to MPE approved Fabric Workspaces